# REGRESSION ANALYSIS (LECTURE 3 )


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**Problem:** For two variables $X$ and $Y$ the two regression lines are given by $4x + y + 8 = 0$ and $x + 6y = 42$. Find the mean of $X$ and $Y$. Also find the correlation coefficient between $X$ and $Y$.

**Answer:** For two variables $X$ and $Y$ the two regression lines are given by $4x + y + 8 = 0$ and $x + 6y = 42$. Solving we get $x = -90/23$ and $y = 176/23$. As two regression lines intersect each other at the point with abscissa as mean of $X$ and ordinate as mean of $Y$ so here mean of $X$ is $-90/23$ and mean of $Y$ is $176/23$. The first equation can be written as $x-\left(-\frac{90}{23}\right)=-\frac{1}{4}\left(y-\frac{176}{23}\right)$

So $b_{xy} = -\frac{1}{4}$. Again, the second equation can be written as $y-\frac{176}{23}=-\frac{1}{6}\left\{x-\left(-\frac{90}{23}\right)\right\}$

So $b_{yx} = -\frac{1}{6}$

Here $b_{yx}.b_{xy}=r^2$. So $r^2 = \left(-\frac{1}{6}\right).\left(-\frac{1}{4}\right)=\frac{1}{24}$

Hence, $r = -\frac{1}{2\sqrt{6}}$

[Here as $b_{yx}$ and $b_{xy}$ are both negative so $r$ is also negative]

**Problem:** Obtain both the regression equations for the given bivariate data:

| | | | | | |
|---|---|---|---|---|---|
| X: | 3 | 2 | 7 | 5 | 8 |
| Y: | 6 | 1 | 8 | 6 | 9 |

Also estimate $y$ for $x = 11$ and $x$ for $y = 10$.

**Answer:** Here we have to find out the two regression lines $Y$ on $X$ and $X$ on $Y$. To estimate $y$ for $x = 11$ we will use regression line of $Y$ on $X$ and to estimate $x$ for $y = 10$ we will use regression line of $X$ on $Y$.

Given bivariate data

| | | | | | |
|---|---|---|---|---|---|
| X | 3 | 2 | 7 | 5 | 8 |
| y | 6 | 1 | 8 | 6 | 9 |

Now here $\bar{x} = (3+2+7+5+8)/5 = 25/5 = 5$

$\bar{y} = (6+1+8+6+9)/5 = 30/5 = 6$

Regression coefficient of y on x $= b_{yx}$

$$= r \cdot \frac{S_y}{S_x}$$

Now $r = \dfrac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})}{S_x \cdot S_y}$

So, $b_{yx} = \dfrac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})}{S_x \cdot \cancel{S_y}} \cdot \dfrac{\cancel{S_y}}{S_x}$

$$= \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})}{S_x^2}$$

Again, $S_x = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2}$. So, $S_x^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2$

Hence, $b_{yx} = \dfrac{\cancel{\frac{1}{N}}\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})}{\cancel{\frac{1}{N}}\sum_{i=1}^{N}(x_i-\bar{x})^2} = \dfrac{\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{N}(x_i-\bar{x})^2}$

Similarly, $b_{xy} = r \cdot \frac{S_x}{S_y} = \frac{\sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N} (y_i - \bar{y})^2}$

| $X$ | $y$ | $x_i - \bar{x}$ | $y_i - \bar{y}$ | $(x_i - \bar{x})^2$ | $(y_i - \bar{y})^2$ | $(x_i - \bar{x})(y_i - \bar{y})$ |
|---|---|---|---|---|---|---|
| 3 | 6 | $3-5 = -2$ | $6-6 = 0$ | 4 | 0 | 0 |
| 2 | 1 | $2-5 = -3$ | $1-6 = -5$ | 9 | 25 | 15 |
| 7 | 8 | $7-5 = 2$ | $8-6 = 2$ | 4 | 4 | 4 |
| 5 | 6 | $5-5 = 0$ | $6-6 = 0$ | 0 | 0 | 0 |
| 8 | 9 | $8-5 = 3$ | $9-6 = 3$ | 9 | 9 | 9 |
| | | | | $\Sigma(x_i - \bar{x})^2 = 26$ | $\Sigma(y_i - \bar{y})^2 = 38$ | $\Sigma(x_i - \bar{x})(y_i - \bar{y}) = 28$ |

Hence, $b_{yx} = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{\Sigma (x_i - \bar{x})^2} = \frac{28}{26} = \frac{14}{13}$

$b_{xy} = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{\Sigma (y_i - \bar{y})^2} = \frac{28}{38} = \frac{14}{19}$

So, equation of regression line of $y$ on $x$:

$y - \bar{y} = b_{yx}(x - \bar{x})$

or, $y - 6 = \frac{14}{13}(x - 5) \Rightarrow 13y - 78 = 14x - 70$

$\Rightarrow \boxed{14x - 13y + 8 = 0}$

Equation of regression line of $x$ on $y$:

$x - \bar{x} = b_{xy}(y - \bar{y}) \Rightarrow x - 5 = \frac{14}{19}(y - 6)$

or, $19x - 95 = 14y - 84 \Rightarrow \boxed{19x - 14y = 11}$

To estimate $y$ for $x = 11$ we use regression line of $Y$ on $X$.

So, we get, $14 \times 11 - 13y + 8 = 0 \Rightarrow 13y = 162$

$\Rightarrow y = \frac{162}{13} = 12.46$

To estimate $x$ for $y = 10$ we use regression line of $X$ on $Y$.

So we get $19x - 14 \times 10 = 11 \Rightarrow 19x = 151 \Rightarrow x = \frac{151}{19}$

$= 7.95$